
```
clc; clear; close all;

% In_class_1a
f = @(x) sin(10*x) + cos(3*x);

x = linspace(3, 6, 1000);
y = f(x);
figure;
plot(x, y, '-', 'LineWidth', 2);
grid on, hold on;

plot([3 6], [0 0], 'r--');
grid on, hold on;

xb = incsearch(f, 3, 6, 100);
[m, n] = size(xb);

% Execute bisection method for each bracket
for i = 1:m
    xL = xb(i,1);
    xU = xb(i,2);
    FL = f(xL);

    xr_old = xL;
    iter = 0;

    fprintf('Root %d Convergence:\n', i)
    fprintf(' iter      xr      err \n')

    while (1)
        xr = (xL + xU)/2;
        FR = f(xr);
        iter = iter + 1;

        err = abs((xr - xr_old)/xr);
        fprintf('%5d %10.4f %10.5f \n', iter, xr, err)

        if err < 1e-4 || iter > 20
            break;
        end

        xr_old = xr;
        if sign(FR*FL) > 0
            xL = xr;
        else
            xU = xr;
        end
    end
end

% Plotting the roots on the curve
plot(xr, FR, 'o', 'LineWidth', 2, ...
```

```

        'MarkerEdgeColor', 'r', ...
        'MarkerFaceColor', 'w', ...
        'MarkerSize', 10);
end
xlabel('x')
ylabel('y')
title('Bisection Method - Incalass 1a')

```

number of brackets:

9

Root 1 Convergence:

iter	xr	err
1	3.2576	0.00465
2	3.2652	0.00232
3	3.2614	0.00116
4	3.2633	0.00058
5	3.2623	0.00029
6	3.2628	0.00015
7	3.2625	0.00007

Root 2 Convergence:

iter	xr	err
1	3.3788	0.00448
2	3.3712	0.00225
3	3.3674	0.00112
4	3.3655	0.00056
5	3.3665	0.00028
6	3.3660	0.00014
7	3.3658	0.00007

Root 3 Convergence:

iter	xr	err
1	3.7424	0.00405
2	3.7500	0.00202
3	3.7462	0.00101
4	3.7443	0.00051
5	3.7453	0.00025
6	3.7457	0.00013
7	3.7460	0.00006

Root 4 Convergence:

iter	xr	err
1	4.2273	0.00358
2	4.2348	0.00179
3	4.2311	0.00090
4	4.2292	0.00045
5	4.2282	0.00022
6	4.2287	0.00011
7	4.2289	0.00006

Root 5 Convergence:

iter	xr	err
1	4.2576	0.00356
2	4.2652	0.00178
3	4.2614	0.00089
4	4.2633	0.00044
5	4.2642	0.00022

6	4.2637	0.00011
7	4.2635	0.00006

Root 6 Convergence:

iter	xr	err
1	4.7121	0.00322
2	4.7197	0.00161
3	4.7159	0.00080
4	4.7140	0.00040
5	4.7131	0.00020
6	4.7126	0.00010
7	4.7124	0.00005

Root 7 Convergence:

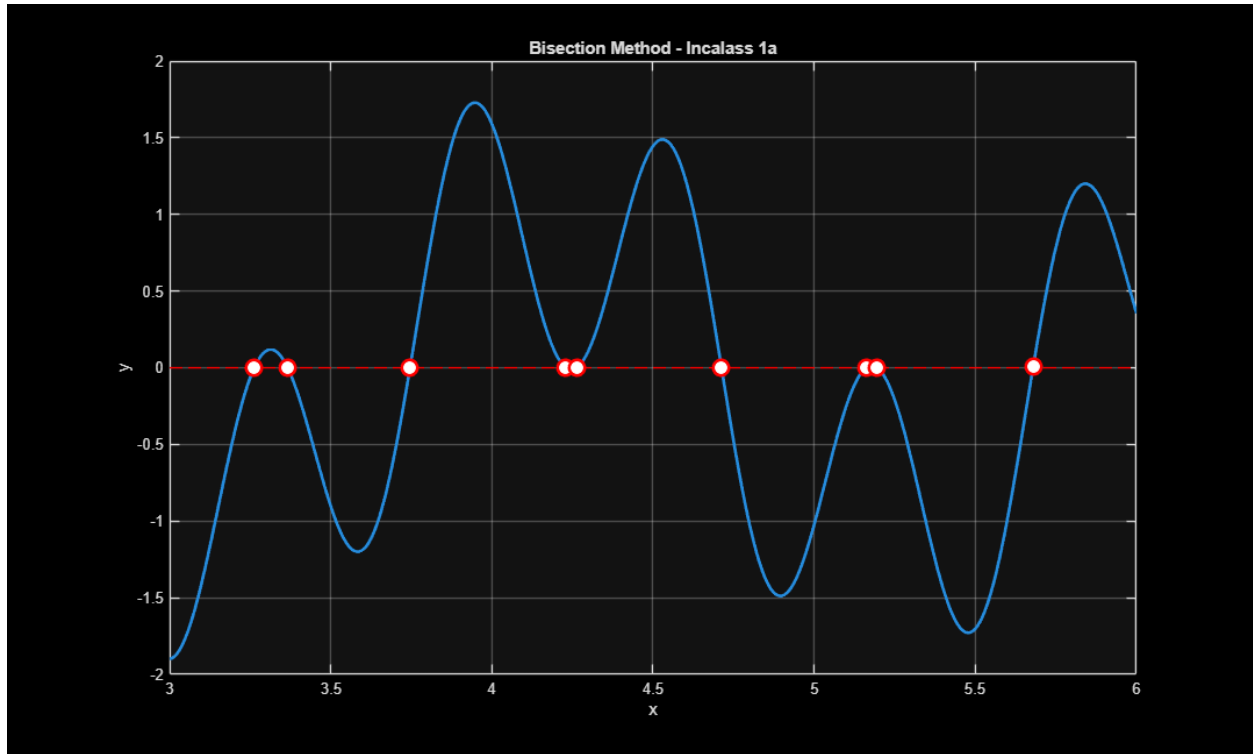
iter	xr	err
1	5.1667	0.00293
2	5.1591	0.00147
3	5.1629	0.00073
4	5.1610	0.00037
5	5.1619	0.00018
6	5.1615	0.00009

Root 8 Convergence:

iter	xr	err
1	5.1970	0.00292
2	5.1894	0.00146
3	5.1932	0.00073
4	5.1951	0.00036
5	5.1960	0.00018
6	5.1955	0.00009

Root 9 Convergence:

iter	xr	err
1	5.6818	0.00267
2	5.6742	0.00134
3	5.6780	0.00067
4	5.6799	0.00033
5	5.6790	0.00017
6	5.6795	0.00008



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